A Quantum Walk Model of Financial Options

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Abstract
Financial markets are often modeled using a random walk, for example in the binomial option pricing model. This paper presents an alternative approach to option pricing based on a quantum walk model. The quantum walk, which incorporates superposition states and allows for effects such as interference, was originally developed in physics, but has also seen application in areas such as cognitive psychology, where it is used to model dynamic decision-making processes. It is shown here that the quantum walk model captures key aspects of investor behavior, while the collapsed state captures the observed behavior of markets. The resulting option price model agrees quite closely with the classical random walk model, but when coupled with a model of supply and demand, helps to explain some observed anomalies. The method also has the advantage that it can be run directly on a quantum device. The aim of this paper is to initiate a discussion about how non-classical models that are native to quantum computers can be applied in finance.

Keywords
financial options, quantum walk, quantum finance, quantum cognition, quantum computing

1. Quantum versus classical walks

One of the key tools in quantitative finance is the random walk model of asset prices. The term "random walk" is attributed to Pearson (1905), who illustrated it using an example of a drunken man walking randomly in an open field; however, the method was actually pioneered in finance by Bachelier (1900), whose doctoral thesis *Theorié de la Spéculation* studied the buying and selling of securities in the Paris Bourse. Bachelier assumed that price fluctuations were random, and the mathematical expectation of profit was zero. The price history of a stock therefore resembled the so-called Brownian motion of a dust particle as it is buffeted around by collisions with individual atoms. Bachelier used this model to derive a method for pricing financial options.

Bachelier's thesis received little attention in his time, however the financial random walk model was rediscovered in the 1950s and was the key inspiration for Eugene Fama's efficient market hypothesis (Fama, 1965), as well as the Black–Scholes model for pricing options (Black and Scholes, 1973). This model revolutionized the world of finance, but its shortcomings have since become increasingly evident (Wilmott and Orrell, 2017), beginning with the 1998 collapse of the investment firm Long-Term Capital Management, whose partners included both Scholes and Merton.

This paper develops a novel and alternative model of financial options based on a quantum version of the random walk. In physics, the idea of a quantum walk goes back at least to Feynman (Wang and Manouchehri, 2013), but was first developed by Aharonov *et al.* (1993). The difference between it and its classical counterpart is that, instead of moving in position space with random steps determined by classical probability, it involves a superposition of states, which evolve in time according to deterministic quantum rules. The system therefore explores multiple possible paths, with probability amplitudes that include the effects of interference.

While the classical random walk involves randomness at each step, the quantum version is deterministic until the final stage when the result is collapsed to a particular outcome. (For this reason it is usually called a quantum walk, rather than a quantum random walk.) The two systems also behave quite differently. In particular, the variance of the quantum walk grows quadratically with time, instead of linearly as in the classical case.

In physics and computing, interest in quantum walks has expanded greatly over the last decade, in part because it was realized that they provide a basis for quantum computation (Childs, 2009). The quantum walk has also found use in psychology for describing a range of cognitive phenomena such as signal detection (Busemeyer and Bruza, 2012), how people assign ratings to stimuli (Wang and Busemeyer, 2016), and general decision-making (Kvam *et al.*, 2015; Martínez-Martínez and Sánchez-Burillo, 2016). Whether in medicine, law, military strategy, business, or any other area, decision-making is a dynamic process where belief states interact with the accumulation of evidence. In a classical model, such as sequential analysis, the decision state is updated each time a new piece of information arrives (Gold and Shadlen, 2007). The noisy nature of the evidence creates a random walk-like trajectory towards one option or the other. In the quantum model, at any time prior to a final decision, the decision-maker is in a superposition of states, with the probability amplitudes of each choice evolving in concert. In terms of neural processes, the quantum walk model corresponds to a massively parallel cognitive architecture that involves both co-operative (excitatory) and competitive (inhibitory) interactions (Fuss and Navarro, 2013).

Cognitive processes are of course very relevant to the pricing of financial options, since this involves an interplay between investor expectations about future asset prices, and the actual asset prices as measured by market transactions. For example, if someone buys a call option it might be because they have a bullish view on the future price, an opinion which will only be subject to final verification at the exercise date. This seems inconsistent with the random walk model, which assumes – as Bachelier did – that the expectation of profit is zero; or Fama's idea that market participants are rational and far-seeing.

In some respects, the task of pricing an option is similar to the psychological experiment, modeled by Kvam *et al.* (2015), using a quantum walk, where participants performed a particular task (judging the motion of dots on a screen) while also being asked to judge the confidence in their decisions. The researchers found that making decisions interfered with the confidence judgments, in such a way as to make the judgments more accurate and less extreme. In the case of financial options, people are being asked to make forecasts about whether the price of the underlying asset will go up or down in the future. It therefore seems reasonable that investors' beliefs about the future may be more extreme than the reality, since they are not subject to measurement in the same way. Indeed, if this weren't the case, and investors thought asset prices would remain stable, then there would be little incentive to purchase options.

At the same time, option prices are the product of supply and demand, and suppliers in particular may tend to use the conventional model to calculate their selling price. As seen below, the quantum walk model can be coupled with a previously published quantum model of supply and demand (see Orrell, 2020a) to accommodate the interplay between these various effects. While quantum methods have previously been applied to the topic of option pricing (Chen, 2004; Baaquie, 2007; Haven and Khrennikov, 2013), efforts have tended to focus on reproducing or modifying classical results such as the Black–Scholes model, while the aim here is to explore non-classical properties such as interference. From a purely technical perspective, the main advantage of the quantum model is that it can run on any device which can simulate a quantum walk, but as we'll see the connection with quantum cognition makes it an informative model in its own right. The outline of the rest of the paper is as follows. Section 2 introduces the mathematical framework for the quantum walk. Section 3 applies this model to the problem of option pricing. Section 4 analyzes the dynamics of supply and demand for the model, and Section 5 summarizes the conclusions.

2. The quantum coin

The quantum walk has been described in detail by a number of authors, so this section gives only a brief overview in order to define some terms used in the option model; see, e.g., Kempe (2003) for more details. Consider first a classical random walk on a one-dimensional grid consisting of regularly spaced points $x_j = j\Delta_x$ for $j \in \mathbb{Z}$ and some spacing Δ_x. The walker starts at initial position $x_0 = 0$, and at each subsequent time point $t_n = n\Delta_t$ tosses a balanced coin, taking a step to the left ($-\Delta_x$) if the coin is heads and a step to the right ($+\Delta_x$) if the coin is tails. After taking n steps, there are 2^n different paths and the final position of the walker will be in the range $-n\Delta_x$ to Δ_x (note that only alternate spaces on the grid can be occupied). In order to determine the probability $P_j(n+1)$ that the walker will end up after the next step in position x_j, we note that the paths to that point must have been at position x_{j-1} or x_{j+1} in the previous step, which gives

$$P_j(n+1) = \frac{1}{2}\left[P_{j+1}(n) + P_{j-1}(n)\right].$$

Carrying this forward a number of steps results in the familiar probability distribution that is described by Pascal's triangle. The variance of the position after n time steps is given by

$$\mathrm{Var}(x) = n\Delta_x{}^2 = \frac{t_n}{\Delta_t}\Delta_x{}^2$$

so increases linearly with time.

In terms of a cognitive model, the random walk model corresponds to the case where a decision-maker makes a random decision (e.g., to move left or right) at each time step. In quantum cognition, we model the mental state of the decision-maker using a quantum wave function, which accounts for objective and subjective factors, and only collapses to a final decision when measured (Busemeyer and Bruza, 2012; see also Orrell, 2020b for a discussion of the applicability of quantum models to economics). It is therefore appropriate to ask how an initial superposed quantum state might evolve by following quantum rules. For example, suppose that you think a stock might go up in price next month. If it does go up in month 1, then in the following month 2 it might also go up or down. Likewise, a down move in month 1 could be followed by either an up or a down move in month 2, and so on. The range of possibilities therefore spreads out over time, as with the classical random walk. But as seen in quantum cognition, the fact that contradictory potential states are held in superposition leads to interference effects, which alter the probability distribution at each time step.

The quantum version of a random walk consists of two parts, a quantum coin and a translation operator. The first plays the role of a coin toss in a classical random walk, while the second determines the consequences of the toss, for example by incrementing the position left or right.[1] The Hilbert space $\mathcal{H}_C$ for a quantum coin toss is spanned by two basis vectors which we identify as $|\uparrow\rangle = \begin{pmatrix}1\\0\end{pmatrix}$ and $|\downarrow\rangle = \begin{pmatrix}0\\1\end{pmatrix}$. The coin toss is represented by a 2×2 Hermitian operator R. Here we will use the so-called Hadamard coin, widely used in quantum computing, which is $R = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}$. It can be shown that all unbiased coin toss operators are essentially equivalent to the Hadamard coin (Kendon and Tregenna, 2003).

Now let the Hilbert space $\mathcal{H}$ be defined as the tensor product $\mathcal{H} = \mathcal{H}_C \otimes \mathcal{H}_P$, where $\mathcal{H}_C$ is the coin space spanned by $\{|\uparrow\rangle, |\downarrow\rangle\}$ and the position space $\mathcal{H}_P$ is spanned by a discrete set of position basis states $\{|x_j\rangle : j \in \mathbb{Z}\}$. We define the translation operator

$$T = |\uparrow\rangle\langle\uparrow| \otimes \sum_j |x_{j+1}\rangle\langle x_j| + |\downarrow\rangle\langle\downarrow| \otimes \sum_j |x_{j-1}\rangle\langle x_j|,$$

which has the effect of shifting the position x_j up (i.e., to the right) or down (to the left) depending on whether the coin state is up or down. A quantum random walk of n steps is defined as the transformation U^n, where U is given by $U = T \cdot (R \otimes I)$. In other words, we apply the coin toss operator R to the current state; apply the transformation operator to shift the position state; and repeat the process n times, with measurement only occurring in the final state.

We can write the wave function at step n as

$$|\psi(n)\rangle = \sum_{j=-\infty}^{\infty}\left(a_j(n)|\uparrow\rangle + b_j(n)|\downarrow\rangle\right) \otimes |x_j\rangle,$$

where $a_j(n)$ and $b_j(n)$ are the coefficients for the up and down components. If we apply the operator U with R as defined above, then the coefficients become

$$a_j(n+1) = \frac{1}{\sqrt{2}}a_{j-1}(n) + \frac{1}{\sqrt{2}}b_{j-1}(n),$$

$$b_j(n+1) = \frac{1}{\sqrt{2}}a_{j+1}(n) - \frac{1}{\sqrt{2}}b_{j+1}(n).$$

The probability distribution of finding that the wave function is either up or down at position x_j can then be found by applying the Born rule, so that

$$P_j(n+1) = \frac{1}{2}\|a_{j-1}(n) + b_{j-1}(n)\|^2 + \frac{1}{2}\|a_{j+1}(n) - b_{j+1}(n)\|^2.$$

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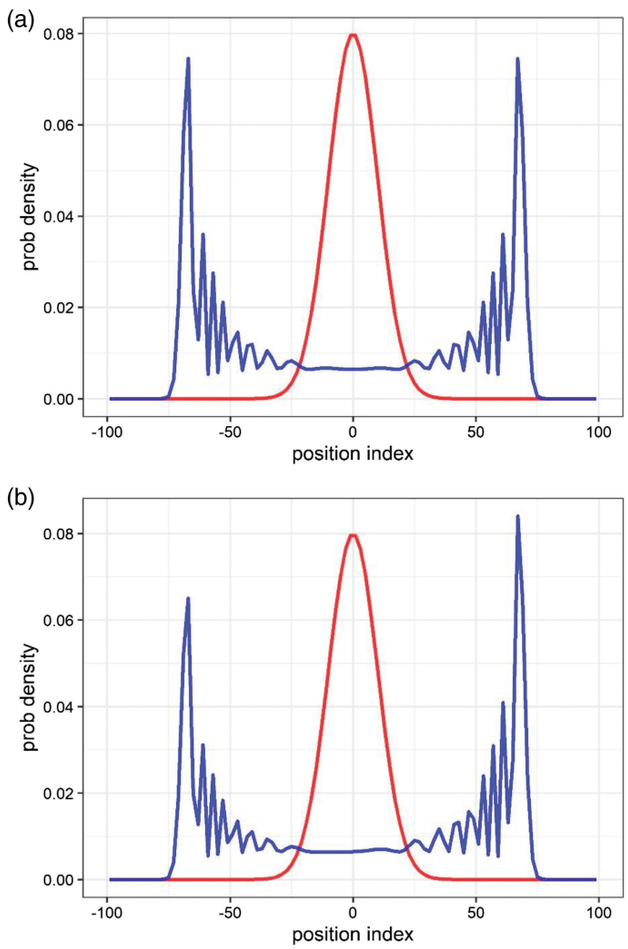
As seen in Figure 1, the result is dependent on the initial state. For the Hadamard coin, setting the initial state to $\psi = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{i}{\sqrt{2}}|\downarrow\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}$ and running the model for 100 steps gives the symmetrical probability distribution shown in the top panel of Figure 1, while a small amount of bias introduces a degree of skew (lower panel, see also Appendix). The shape resembles the wave function of a quantum harmonic oscillator in an excited state and can be compared with the propensity curves in Orrell (2020c), which represent the price distribution for a potential transaction. The irregular and jagged nature of the plot may seem strange, but as with the quantum oscillator, what counts at the macro level is the smoothed behavior. Another difference from the classical model is that the variance of position increases in a quadratic fashion (Chandrashekar *et al.*, 2008), and for large n is given by

$$\mathrm{Var}\,(x) \cong \left(1 - \frac{1}{\sqrt{2}}\right) n^2 \Delta_x{}^2 = \left(1 - \frac{1}{\sqrt{2}}\right)\left(\frac{t_n}{\Delta_t}\right)^2 \Delta_x{}^2.$$

Figure 1: Blue lines show probability distribution for a quantum random walk of 100 steps using the Hadamard coin, moving on a 1-D grid with $\Delta_x = 1$. Top panel has a balanced initial state, lower panel has a small amount of bias toward higher prices. Red lines show the classical binomial distribution.



Between the cases with perfect interference and no interference (i.e., classical), there is also the intermediate case where there is a degree of interference, due for example to the effect of decoherence through entanglement with the environment. In this case the variance can be shown to be asymptotically linear (Brun *et al.*, 2003). Decoherence can be added in a number of ways. Perhaps the simplest is to slightly perturb the coin toss operator at each step. Figure 2 shows an ensemble average of 400 simulations, where the coin toss operator was an imperfect Hadamard

$$R_n = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & e^{i\varsigma}\\ e^{-i\varsigma} & -1\end{pmatrix},$$

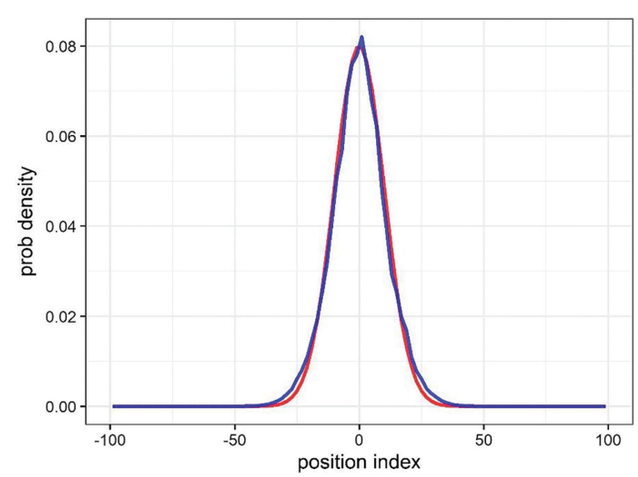
with the parameter ζ randomly drawn at each time step n from a normal distribution with mean 0 and standard deviation $\sigma_{dec} = 1$ (Mackay *et al.*, 2002; Chandrashekar *et al.*, 2008). If the system is measured at each time step, then decoherence is complete and the quantum model collapses to the classical case.

To summarize, when used as a model of cognition, the quantum walk has a number of characteristics which distinguish it from classical random walk models. One is that the system has a degree of momentum, so the initial condition determines whether the distribution is balanced or skewed. Variance of the system grows quadratically rather than linearly, which again is consistent with the idea of a linear momentum pushing to one side or the other. Finally, the system is sensitive to decoherence, and collapses to the classical case if measurement occurs at each time step. The next section shows how the quantum walk can be used to model not stock prices per se, but rather subjective opinions about future stock prices, which are not the same thing, and thus arrive at a model for pricing options.

3. Quantum binomial asset price model

The classical binomial option-pricing model was developed by Cox *et al.* (1979), and exists in different versions. The one we will use here is based on the Jarrow–Rudd "equal probabilities" model, where the probability of a move up or down is the same (Jarrow and Rudd, 1983). In the usual model, the steps up and down in log price are given by

Figure 2: Blue line shows the ensemble average of 400 simulations for the imperfect Hadamard coin with decoherence $\sigma_{dec} = 1$. Each simulation is over 100 steps. Red line shows the classical case.



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$$u = \left(r - \frac{\sigma^2}{2}\right)\Delta_t + \sigma\sqrt{\Delta_t},$$

$$d = \left(r - \frac{\sigma^2}{2}\right)\Delta_t - \sigma\sqrt{\Delta_t},$$

where r is the risk-free return, and σ is the asset's volatility (defined as the annualized standard deviation of the continuously compounded return).[2] Because we assume a fixed step size, we interpret this as a random walk with $u = \sigma\sqrt{\Delta_t}$ and $d = -\sigma\sqrt{\Delta_t}$, in combination with a drift term $r\Delta_t$ due to the risk-free interest rate, and a small adjustment term $-\frac{\sigma^2}{2}\Delta_t$.

In the quantum version, we similarly interpret the asset price evolution as a quantum walk on a grid with an additional drift (Orrell, 2020d). Because the quantum model diverges more quickly than the classical model, we also need to adjust the grid size in order to give the desired variance at the final time. Alternatively, the time step can be scaled to give the same final variance, which gives a significant speed boost over the classical method, as in quantum search algorithms (Grover, 1996).

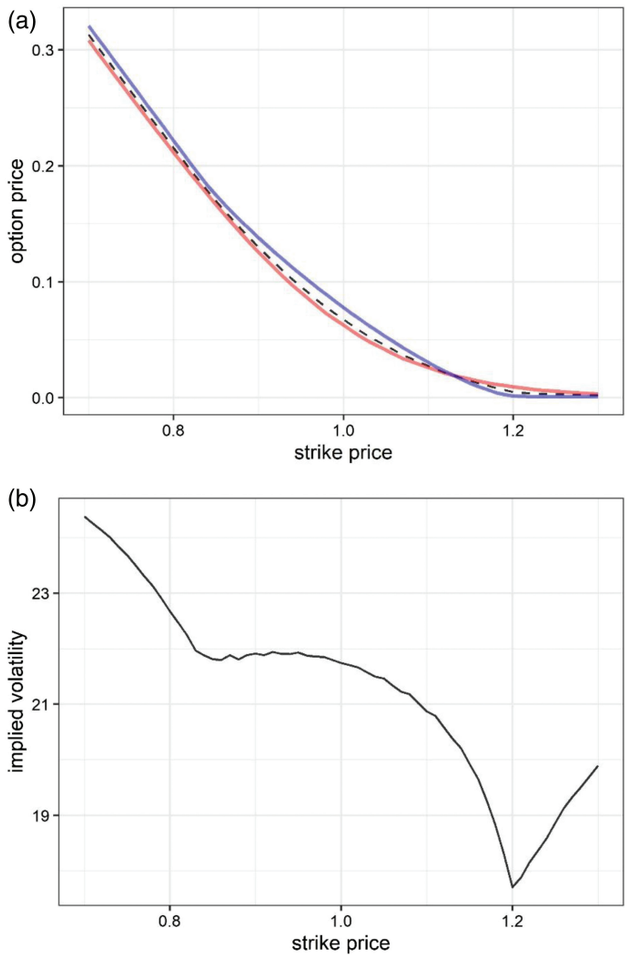
Now, consider a European-style call option with strike price K. After n time steps, the expected value of the payoff is $\left[\exp\left(S_n\right) - K\right]^+$ (the exponential is required because the stock price S_n is logarithmic). The option price V_n is the expected payoff discounted to time zero, or

$$\langle V_n \rangle = \left\langle \frac{1}{(1+r)^n}\left[\exp\left(S_n\right) - K\right]^+ \right\rangle. \tag{1}$$

This can be computed directly for the quantum or classical cases from the probability distributions for the stock. The algorithm is therefore as follows: perform a quantum walk over a position index; interpret position as a log price; adjust the price increment to give the desired volatility; shift to account for the risk-free interest rate; and use the final probability distribution for the log price to compute the option price using the formula in Eq. (1).

The top panel of Figure 3 shows the computed option price for different values of strike price, where the initial purchase price is 1 and maturity time is 6 months, for the quantum and classical cases. The quantum walk here includes a small amount of bias towards positive prices, as in the lower panel of Figure 1. Note that the quantum model can be brought as close as desired to the classical model by adding a degree of decoherence. This decoherence can be used as a way to simulate the effect of information, based for example on historical option prices or the opinions of other market participants, which brings investor projections about the future closer to the classical model.

Figure 3: Top panel shows the market option price (dashed line), which is a compromise between the quantum (blue) and classical (red) models, for an initial purchase price of 1, annualized volatility 20 percent, risk-free interest rate 2 percent, and maturity time 6 months. The market model incorporates the effects of supply and demand and is constructed as described in the Appendix. Lower panel shows a plot of implied volatility versus strike price.



4. Supply and demand

A key distinction between modeling the price of an option and the price of the underlying stock is that the option concerns not the actual price evolution of the stock, but rather its imagined possible future evolution. In the classical random walk model these are treated as the same thing, which leads to two paradoxes. Firstly, investors would have little motivation to purchase options, because there would be no expected profit or need to hedge against extreme price changes. Secondly, buyers and sellers would use the same model, so prices should follow the model exactly, and (apart from behavioral factors) volume should be independent of strike price. In reality, investors often purchase options because they anticipate significant price changes; prices depart from the classical model, as shown by phenomena such as the volatility smile or skew; and volume is not uniform, but is concentrated in a particular range. These effects can only be explained by assuming that buyers and sellers are using different models or approaches, with prices emerging as the result of a negotiation.

A general method for modeling the price negotiation process was presented in Orrell (2020b,c). The price is described by a propensity function, which is a normal probability distribution with mean

$$\mu = \frac{\sigma_b^2\mu_s + \sigma_s^2\mu_b}{\sigma_t^2}$$

and standard deviation

$$\sigma = \frac{\sigma_s\sigma_b}{\sigma_t},$$

where $\sigma_t = \sqrt{\sigma_s^2 + \sigma_b^2}$. Here, μ_s and μ_b are the ideal prices for the seller and the buyer, respectively, while σ_s and σ_b are a measure of price flexibility. The transaction volume scales with

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$$\alpha = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{\mu_t^2}{2\sigma_t^2}\right),$$

where $\mu_t = \mu_s - \mu_b$.

A first step to model option pricing would be to assume that sellers price options according to the dominant classical model price p_c, while buyers price options according to the quantum model price p_q, which takes into account subjective projections about the future. If we allow for a margin on each side, then the ideal prices for buyer and seller are

$$\mu_s = p_c + d_s,$$

$$\mu_b = p_q - d_b,$$

where d_s is the margin for the seller and d_b is the margin for the buyer. For simplicity, we will assume in the code that the relative size of the margins reflects the price flexibility, so that a smaller margin implies less flexibility: $\mu_s / \mu_b = d_s / d_b$. By assigning values for the margins, and an overall price flexibility σ_t, we can then solve for the terms σ_s and σ_b, which reduces the number of free parameters. This also means that if the classical and quantum models agree on price, then changing the margins has no effect on expected price, and in general the expected price is quite insensitive to the choices of margin. For the parameters in the Appendix, the model assigns a larger margin to the buyer than to the seller, which means the seller has more control over the price (in terms of the model parameter interpretation, they have a greater mass in the transaction).

The resulting market price, shown by the dashed line in the top panel of Figure 3 for a 6-month option, is therefore a kind of compromise between the quantum and classical models. The lower panel of Figure 3 shows a plot of the resulting implied volatility. The curve is found by taking the market price of the option at each value of strike price, and using an optimization routine to compute the volatility which would produce that price using the classical binomial model (which is nearly indistinguishable from the Black–Scholes model). It can be compared with Figure 4, which is a surface plot showing implied volatility for a range of expiration dates up to 1 year. The resulting surface resembles the implied volatility surfaces obtained from empirical analysis of option price data (see, e.g., Cont and Fonseca, 2002); however, analysis is complicated by the fact that a number of other factors also contribute, including behavioral effects such as loss aversion (Versluis *et al.*, 2010; Wilmott, 2013). One advantage of the quantum model is that it can be integrated with models from quantum decision theory to account for such effects (see, e.g., Yukalov and Sornette, 2018).

The most noticeable difference between the quantum and classical models is that the quantum price is higher for at-the-money options (i.e., options with a strike price near 1). In the model, this translates into higher volume for that range of strike prices, as shown by the surface plot in Figure 5 for options with expiration dates up to 1 year, and volatility here set to 40 percent. Again, this agrees with the empirical observation that volume is highest for at-the-money options. For example, Bergsma *et al.* (2019) examined data for US exchange-traded options over the period 2006–2017. The options studied were between 10 and 60 days to expiration, and had a mean implied volatility of 39.5 percent. They found the mean "moneyness" or normalized strike price was 0.988, with a standard deviation of 0.057. Pan and Poteshman (2006) studied a data set of Chicago Board Options Exchange-listed options from 1990 to 2001, and reported a similar distribution but with a standard deviation of about 0.1. The larger standard deviation is probably because the data included options with longer expiration dates.

One unrealistic feature of the quantum model is that it tends to assign very low probability to the purchase of options that are far out-of-the-money. According to Bergsma *et al.* (2019), "informed traders" – who are presumably more objective about prices – often prefer to use options that are out-of-the-money because they offer higher leverage. In contrast, less sophisticated traders may be attracted by the opportunity to make lottery-style bets on big price changes. Such effects could be accounted for in a more complicated model by including a population of purchasers who use the classical model, rather than the quantum model; or by modifying the purchase propensity function to include behavioral effects.

The quantum walk model may also help to explain the empirically noted property that the implied volatility includes a time-dependent term that varies inversely with the square root of the time to expiry (Fouque *et al.*, 2004; Daglish *et al.*, 2007).

Figure 4: Implied volatility surface for the market option price model as a function of strike price and expiration, for a call option with initial purchase price of 1, annualized volatility 20 percent, and risk-free interest rate 2 percent.

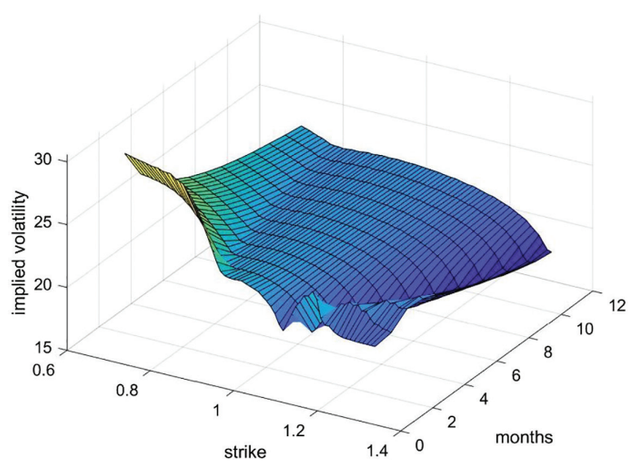
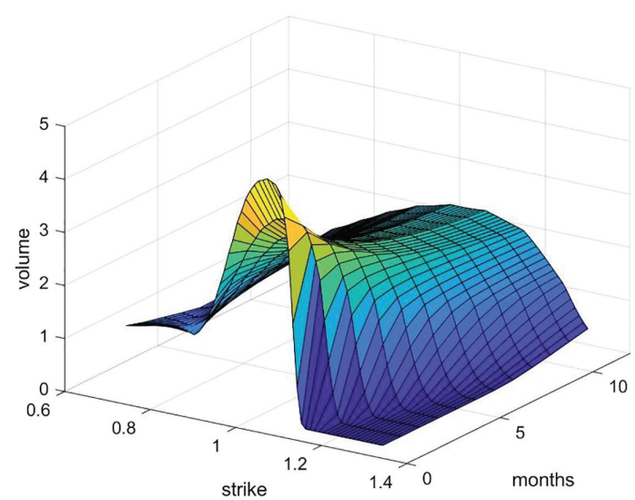


Figure 5: Probability distribution surface for the volume of transactions as a function of strike price, for an option with volatility of 40 percent and expiration dates up to 1 year. The surface is normalized at each time point, so that the total probability for any time is 1.



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Because the quantum volatility scales linearly with time, but the observed volatility scales with the square root of time, it follows that, from the point of view of the investor, asset fluctuations which follow a classical random walk with constant volatility will imply a higher internal (i.e., quantum) volatility when viewed over a period of say 1 month than when viewed over a longer period such as 6 months. The incompatibility between internal (subjective) and measured (objective) growth rates, which has a square root of time dependency, may therefore manifest itself as an adjustment to option prices (not modeled).

While a full empirical analysis of market price data is beyond the scope of this paper, these results suggest that the quantum model may be useful for explaining some of the departures of investor behavior from the classical model. Of course, the main advantage of the quantum walk model, from the standpoint of quantum computation, is that it can be implemented directly on a quantum computer. Such computers, if they become fully operational, will offer significant speed advantages for certain classes of problem (Rebentrost *et al.*, 2018; Orús *et al.*, 2019). Since the classical model can be computed by running the quantum model with decoherence, as explained above, it follows that a quantum computer – or more generally, a device that can perform a quantum walk – could simulate both the classical model and the expected departure from that model due to quantum cognition effects.

5. Quantum-native finance

To summarize, the main conclusion of the paper is that the quantum walk model offers a plausible alternative to the random walk traditionally used in finance, which offers the following advantages:

- The quantum walk model captures key aspects of investor behavior and reflects investors' subjective beliefs in the absence of confirming information.
- Adding decoherence to the quantum model brings it arbitrarily close to the classical model, and can be used as a way to reflect the role of external information.
- When integrated with a quantum model of financial transactions, the model captures many aspects of empirical option data, in terms of both the volatility surface and the transaction volume.
- The model can be implemented directly on a quantum computer or device.

From the standpoint of traditional practice, it might seem unrealistic to use a model for stock prices which (at least for the case with no decoherence) departs so greatly from the standard Gaussian distribution used in classical random walks. However, it is again important to note that what is being modeled is not stock price movements, but rather ideas held by investors about future stock price movements. While it is certainly possible and appropriate to model stock prices using a quantum formalism, the price wave function is collapsed every time there is a transaction (although transactions may be infrequent; see Sarkissian, 2020). With options, this collapse doesn't occur in the same way, just because the prices are hypothetical. One would therefore expect investors' ideas about prices to be more extreme and divided than the prices themselves.

Another difference between the classical and quantum models is that classical option prices can be justified by using a hedging argument. The proof of the Black–Scholes model, for example, was based on constructing a perfectly hedged portfolio consisting of the option and the underlying stock. However, hedging does not play the same role in the quantum model, because the wave function is only collapsed at the final step. Measuring the price at each time step would continually collapse the wave function and make it the same as the classical case. A topic for future work is to consider how the theory can apply to portfolios of stocks and options.

In its base form, the quantum model is parsimonious and introduces no extra parameters, other than the degree of decoherence if desired. The quantum model also provides an interface to incorporate other quantum models of cognition and group behavior, in order to simulate, for example, the effects on investor optimism or entanglement between agents. It is hoped that this paper will help to initiate a more general discussion about how non-classical models that are native to quantum computers can be applied in finance.

Appendix

The model used to simulate the sale and purchase of options is the quantum model of supply and demand, which is described in detail in Orrell (2020c,d).

The seller's optimal price is determined by running the classical binomial model and adding a margin of 10 percent. The buyer's optimal price is based on the quantum model, plus a margin of 20 percent. Note that these margins represent the difference between the model price and an optimal desired price, so are larger than something like a profit margin; however, the model is less sensitive to their magnitudes than to their proportionate levels. Both prices are also increased by a small amount of 0.001, which puts a floor on the prices of cheap out-of-the-money options to reflect a minimal transaction cost. The start angle of the quantum model can also be adjusted if desired (neutral is 45 degrees). Here it is set to 50 degrees, which gives a slight bias toward higher prices on the part of the buyer, and introduces a degree of skew to the implied volatility curve. The remaining parameter is a total standard deviation σ_t = 0.15 for the transaction price, which is assumed to be constant and affects only the volume calculation.

The resulting model can be viewed as a market compromise between the quantum and classical models, which are shown in the top panel of Figure 3. Note that results such as the shape of the volatility and volume surfaces in Figures 4 and 5 are not highly sensitive to the choices of parameters for the supply/demand model, because they are caused by the difference between the quantum and classical option pricing models.

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The figures in the paper were produced using the QuantumOption app, available at https://david-systemsforecasting.shinyapps.io/QuantumOption/

David Orrell is an applied mathematician and writer. His books include *The Money Formula: Dodgy Finance, Pseudo Science, and How Mathematicians Took Over the Markets* (with Paul Wilmott) and *Quantum Economics and Finance: An Applied Mathematics Introduction*. He lives in Toronto.

Endnotes

1. This method compares with that of Chen (2004), who treated the two operations in a single step, in order to re-derive the classical binomial option pricing model using a quantum model.
2. The factor 2 arises because only alternate points on the grid are occupied.

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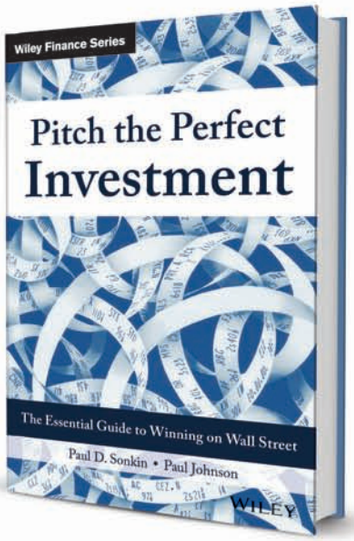

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